What Target-wise ABA Learning Does on Deterministic Causal Fixtures

Evidence from controlled probes H0–H7b

Working synthesis for discussion with Fabrizio

4 August 2026

Status and purpose. This is a supervisor-facing working document. It introduces the causal fixture, learner input, terminology, configurations, and experimental sequence before stating any cross-cutting findings. The controlled probes labelled H0–H7b are complete and analysed; the labels are internal shorthand only and each probe is described in the text by its scientific question and design. No cross-cutting conclusions are asserted yet. The six Findings headings and the Implications heading at the end are intentionally empty placeholders for subsequent drafting and review.

Abstract

The investigation asks what target-wise ABA Learning does when given only a finite binary table sampled from a known causal data-generating process. The principal process is a binary collider $A \to C \leftarrow B$ with independent stochastic roots and a deterministic AND child $C = A \wedge B$. The graph, mechanisms, population distribution, finite sample, learner-visible encoding, learned ABA framework, and evaluator-only reference are kept separate throughout.

The experimental sequence is recorded under labels H0–H7b for reference. It begins with a baseline comparison of the published Greedy and non-deterministic configurations across all targets on one frozen table (H0), then introduces one controlled question at a time: finite-sample support (H1); an irrelevant trailing covariate (H2); background-knowledge order with a leading distractor (H3); brave versus cautious learning (H4–H5); folding-token budget and grounded sample size (H6); and the `relto` versus `sechk` assumption-introduction options (H7). This document records what each probe was intended to reveal, its principal signal, and the procedural explanation supported by the generated deltas and Prolog traces.

Contents

1 Purpose, scope, and organising question

1.1 Research question

> For a finite binary table generated by independent stochastic roots and deterministic non-root mechanisms, and for each observed variable taken in turn as the learning target: what ABA framework does ABALearn emit under the configurations studied here, and how does that framework relate to the evaluator-only mechanism reference and to the information present in the population and in the sample?

The purpose is to describe how the learner transforms the supplied table into rules, assumptions, and contraries, and to classify each emitted theory against a target-dependent evaluator reference. On the principal AND collider $A \to C \leftarrow B$, only the deterministic child C has a desired learned rule over its observed parents:

```
c(A) :- a_val_1(A), b_val_1(A).
```

The roots A and B are generated by independent stochastic mechanisms. There is no underlying deterministic rule relating either root to the other observed variables, and none should be learned.

Relative to that target-dependent reference, the outcomes distinguished in this document are:

- for a deterministic child: a rule that coincides with the mechanism;
- for a deterministic child: a sample-adequate but non-minimal rule;
- for a deterministic child: a defeasible representation that differs from the mechanism string;
- for a root or child: a rule that holds on the finite sample because a population counterexample is missing from that sample;
- for a root: an assumption construction that satisfies the selected learning semantics while leaving the target underdetermined by its predictors;
- search termination under the configured procedure, including timeout or `completed_no_solution`, which for a stochastic root target may be the intended outcome under the reference above.

These classes separate procedural exit status, finite-sample adequacy, and mechanism alignment under the target-dependent reference.

1.2 Scope

The evidence comes from three small binary fixtures, fixed seed-42 samples, and the learner configurations described below. Learning is target-wise: one frozen table is held fixed, and each observed variable is selected in turn as the learning target. Changing the target changes the positive and negative examples and the background-knowledge encoding of the remaining columns; the underlying rows remain the same sample.

Only the deterministic child has an evaluator-desired deterministic rule over its observed parents. Root-target runs are diagnostics of the same pipeline when the target itself is generated stochastically: they show how finite support, learning semantics, and search budget shape the emitted theory or search outcome when no underlying deterministic rule from the other variables should be learned.

1.3 Object of study

The pipeline under study is target-wise ABA Learning over encoded tabular data. For each target it receives background knowledge and labelled examples only, and it emits an ABA framework of rules, assumptions, and contraries under a selected learning semantics and search configuration.

The evidence recorded here therefore concerns learned rule structure, search behaviour, and the relation of those outputs to the evaluator-only mechanism reference for the fixture under study.

2 Conceptual framework and notation

This section fixes the objects, ABA notions, target-wise encoding, and learner controls used throughout the experimental sequence.

2.1 Seven objects kept separate

The experimental reasoning distinguishes the following objects:

1. a generating DAG G;
2. structural mechanisms for the variables in G;
3. the exact observational population distribution P^*;
4. a finite IID sample $\mathcal{D}_n \sim P^*$;
5. target-specific learner input derived from $\mathcal{D}_n$;
6. the learned ABA framework and its emitted delta (the learned additions relative to the input background knowledge); and
7. evaluator-side judgements about sample adequacy, mechanism alignment, and minimality relative to the target-dependent reference.

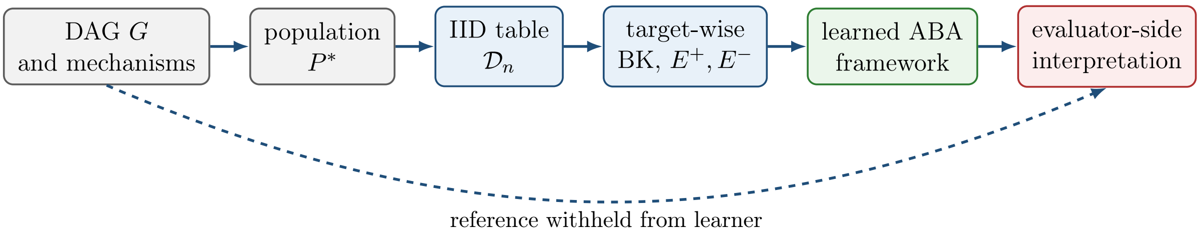

Figure 1: Experimental separation. Only the finite table and its target-wise encoding enter the learner. The generating graph and mechanism reference are reserved for evaluation.

Ordinary faithfulness is a property of the pair (P^*, G). Finite-sample adequacy is a property of a learned framework relative to $\mathcal{D}_n$ under the selected learning semantics. A parent literal appearing in a learned body is a syntactic fact about the emitted delta; any judgement that the delta aligns with the mechanism is an evaluator-side reading.

2.2 ABA framework

An ABA framework is a tuple

$$\mathcal{F} = \langle \mathcal{L}, \mathcal{R}, \mathcal{A}, \bar{\cdot} \rangle,$$

where $\mathcal{L}$ is a language, $\mathcal{R}$ is a set of rules, $\mathcal{A} \subseteq \mathcal{L}$ is a set of assumptions, and $\bar{\cdot}$ maps each assumption $\alpha \in \mathcal{A}$ to its contrary $\overline{\alpha} \in \mathcal{L}$.

A rule has the form

$$h \leftarrow b_1, \ldots, b_m.$$

If $m = 0$, it is a fact. Write $\Delta \vdash_{\mathcal{R}} s$ when a claim $s \in \mathcal{L}$ is derivable from a set of assumptions $\Delta \subseteq \mathcal{A}$ using the rules in $\mathcal{R}$.

Attacks are generated by contraries: Δ attacks an assumption $\alpha \in \mathcal{A}$ when $\Delta \vdash_{\mathcal{R}} \overline{\alpha}$. A set $\Delta \subseteq \mathcal{A}$ is *conflict-free* if it attacks none of its own members. It is *closed* if whenever $\Delta \vdash_{\mathcal{R}} \alpha$ for some $\alpha \in \mathcal{A}$, then $\alpha \in \Delta$.

A set $\Delta \subseteq \mathcal{A}$ is a *stable extension* of $\mathcal{F}$ if it is closed, conflict-free, and attacks every assumption in $\mathcal{A} \setminus \Delta$. Write $\mathrm{Stab}(\mathcal{F})$ for the collection of all stable extensions of $\mathcal{F}$.

A claim $s \in \mathcal{L}$ is accepted with respect to a stable extension Δ if there exists $\Delta' \subseteq \Delta$ such that $\Delta' \vdash_{\mathcal{R}} s$; write $\mathcal{F} \models_{\Delta} s$ when this holds. Write $\mathcal{F} \models_{\text{caut}} s$ when s is accepted with respect to every stable extension of $\mathcal{F}$. A learned framework may therefore combine ordinary rules with explicit exceptional conditions expressed by assumptions and contraries.

2.3 ABA Learning task

An ABA Learning task begins with background knowledge and two finite sets of examples:

$$E^+ = \{e_1^+, \ldots, e_p^+\}, \qquad E^- = \{e_1^-, \ldots, e_q^-\}.$$

The learner applies symbolic transformations to construct a framework $\mathcal{F}'$ under which the examples satisfy the selected consequence condition. The learned object is itself an ABA framework.

The transformations relevant here are:

- **Rote learning.** Add example-specific rules from the positive examples.
- **Folding.** Replace example-specific conditions with reusable BK predicates, producing intensional rules.
- **Subsumption.** Remove a specialised rule when an appropriate more general rule already exists.
- **Assumption introduction.** Make an over-general rule defeasible by adding an assumption, then learn rules for its contrary to represent exceptions.

2.4 Exact-value target-wise encoding

In this investigation the learning task is instantiated from a tabular sample. For each row identifier i, every non-target variable is represented by exactly one value predicate. When c is the target, for example, the BK contains facts such as

```
a_val_1(9).
b_val_0(9).
```

and row 9 contributes either $c(9) \in E^+$ or $c(9) \in E^-$, according to its target value. For a binary target Y,

$$E_Y^+ = \{Y(i) : Y_i = 1\}, \qquad E_Y^- = \{Y(i) : Y_i = 0\}.$$

All non-target columns are available as BK predictors in their serialised order. The encoding interface is the same for every target. The evaluator-side desired object still depends on the target's generating role: only a deterministic child has a desired parent mechanism rule.

2.5 Brave and cautious learning

Brave and cautious learning are the two consequence conditions under which a learned framework $\mathcal{F}'$ may be required to cover E^+ and exclude E^-.

Brave learning requires an existential witness: there must exist some $\Delta \in \mathrm{Stab}(\mathcal{F}')$ that simultaneously accepts every positive example and rejects every negative example,

$$\exists \Delta \in \mathrm{Stab}(\mathcal{F}') \quad \big[\forall e \in E^+,\ \mathcal{F}' \models_{\Delta} e \ \wedge\ \forall e \in E^-,\ \mathcal{F}' \not\models_{\Delta} e\big].$$

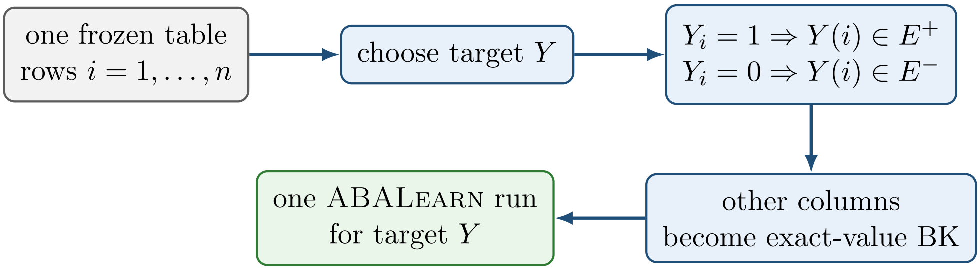

Figure 2: Target-wise encoding. All targets use the same rows; the selected target determines the positive and negative examples and the remaining columns determine the BK.

Cautious learning requires every positive example to be a cautious consequence and every negative example to be excluded as a cautious consequence,

$$\forall e \in E^+,\ \mathcal{F}' \models_{\text{caut}} e, \qquad \forall e \in E^-,\ \mathcal{F}' \not\models_{\text{caut}} e.$$

Thus the two modes quantify differently over $\mathrm{Stab}(\mathcal{F}')$: brave learning uses one witnessing extension, while cautious learning uses acceptance in every stable extension.

2.6 Greedy and non-deterministic folding

Folding is controlled by `folding_mode`, together with companion options for selection and folding space. The two modes used here are as follows.

`folding_mode(greedy)`. A rote rule is folded in one deterministic pass against matching background predicates, under `folding_selection(mgr)` and `folding_space(bk)`. This is the folding regime of the AAMAS configuration.

`folding_mode(nd)`. Folding proceeds incrementally under a folding-token budget (`folding_steps`), with `folding_selection(any)` and `folding_space(all)`. This is the folding regime of the ECAI configuration and of the cautious nd configurations used later.

Each published configuration also sets learning semantics and `asm_intro`. When two named configurations are compared as wholes, several options change together.

2.7 Assumption introduction: `relto` and `sechk`

Assumption introduction is the transformation that makes an over-general folded rule defeasible by placing an assumption in its body and learning rules for the contrary of that assumption. In ABALEARN it is controlled by the option `asm_intro`, which takes one of two values: `relto` or `sechk`. Both are used after a folded candidate fails the configured post-folding entailment check. They differ in how the first assumption-repair step is chosen.

`asm_intro(relto)`. The learner first searches the current framework for an existing assumption that is relative to the body of the folded rule under consideration. If such an assumption is found, it is reused in the repaired rule. If none is found, the learner may generate a new assumption and continue along the subsequent generalisation path.

`asm_intro(sechk)`. The learner first generates a new provisional assumption for the folded rule and then subjects the resulting framework to the semantic-checking branch of generalisation (`gen5`/`gen6`), which inspects stable-extension consequences before deciding how the assumption is retained or revised.

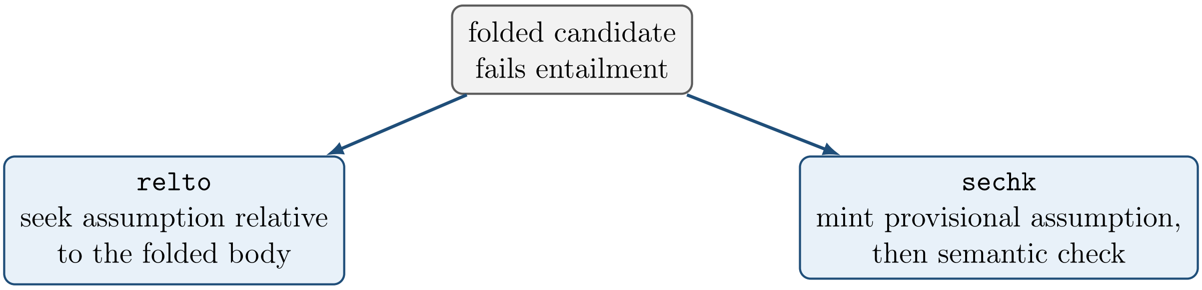

Figure 3: The two `asm_intro` options as alternative first steps of assumption introduction.

3 Causal fixtures and evaluator ground truth

This section records the generating processes, population certificates, evaluator-only mechanism references, and frozen samples used by the probes below. Mathematical labels are A, B, C, D; learner-safe identifiers are `a`,`b`,`c`,`d`. In emitted Prolog rules the capital argument (for example the `A` in `c(A)`) is the row identifier, not the causal variable A.

3.1 Principal fixture: binary AND collider (H0)

The principal fixture is the unshielded collider

$$G_0 : \qquad A \longrightarrow C \longleftarrow B$$

with no A–B edge. The roots are mutually independent,

$$A \sim \mathrm{Bernoulli}(4/5), \qquad B \sim \mathrm{Bernoulli}(7/10), \qquad A \perp\!\!\!\perp B,$$

and the child is the deterministic AND mechanism

$$C = A \wedge B.$$

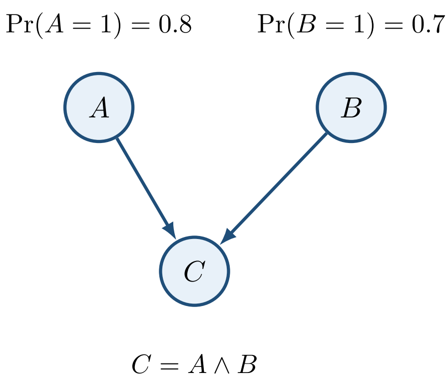

Figure 4: Principal generating DAG (H0). Randomness enters through the independent roots; the non-root mechanism is deterministic.

The induced observational population has four positive-probability atoms:

A	B	C	$P^*(A,B,C)$
0	0	0	$3/50 = 0.06$
0	1	0	$7/50 = 0.14$
1	0	0	$6/25 = 0.24$
1	1	1	$14/25 = 0.56$

The remaining four binary assignments $(0,0,1)$, $(0,1,1)$, $(1,0,1)$, and $(1,1,0)$ are structural zeros.

Causal sufficiency is declared by fixture design: the root exogenous noise terms are mutually independent and there is no omitted common cause. The causal Markov condition holds by construction of the joint,

$$P^*(a,b,c) = P^*(a)P^*(b)P^*(c \mid a,b).$$

Ordinary faithfulness was verified exactly by auditing every singleton-pair conditional-independence query against d-separation. The sole population independence is $A \perp\!\!\!\perp B$; conditioning on the collider yields $A \not\perp\!\!\!\perp B \mid C$, and each parent is dependent on C.

Under ordinary conditional-independence semantics, the skeleton is $A - C - B$ and the unshielded collider $A \to C \leftarrow B$ is oriented in the CPDAG; the corresponding MEC contains only G_0. This is the ordinary CI-based reading. Deterministic mechanisms can impose additional functional constraints beyond ordinary CI; no extended recovery object is asserted here. The graph, certificates, and mechanism reference are evaluator-only and are withheld from ABALearn.

3.2 Evaluator-only mechanism reference

For the deterministic child C, the canonical positive-state exact-value rule is

```
c(A) :- a_val_1(A), b_val_1(A).
```

On this fixture the formal truth-table rule, the population-supported rule, and the rule supported by the frozen sample coincide on that string. That string is the evaluator reference against which learned theories for target `c` are compared. Comparison uses rule bodies, assumptions, contraries, and the target's generating role together: a mechanism-aligned string is one form of evidence; an assumption-rich but sample-adequate theory is another object under the same reference.

The roots A and B are generated by independent stochastic mechanisms $A \sim \text{Bernoulli}(4/5)$ and $B \sim \text{Bernoulli}(7/10)$. Neither root is a deterministic function of the other observed variables, so when `a` or `b` is the learning target no underlying deterministic rule from those predictors should be learned.

3.3 Frozen sample for the principal fixture

The principal table is the target-free IID sample with $n = 30$ and seed 42:

Assignment	Count
$(1,1,1)$	17
$(1,0,0)$	7
$(0,1,0)$	4
$(0,0,0)$	2

All four population atoms appear. Under the binary exact-value encoding the implied example counts are

$$a : 24^+/6^-, \qquad b : 21^+/9^-, \qquad c : 17^+/13^-.$$

For child target `c`, every positive example has $(A, B) = (1, 1)$. For root target `a`, the predictor cell $(B, C) = (0, 0)$ contains both labels:

Row	A (target)	B	C	Class for target `a`
9	1	0	0	positive
26	0	0	0	negative

Rows 9 and 26 therefore share the same BK literals `b_val_0` and `c_val_0` while carrying opposite labels for A. No deterministic rule over (B, C) separates them. Target `b` has the symmetric ambiguity on predictor cells involving (A, C).

3.4 Controlled extensions: trailing and leading distractors (H2, H3)

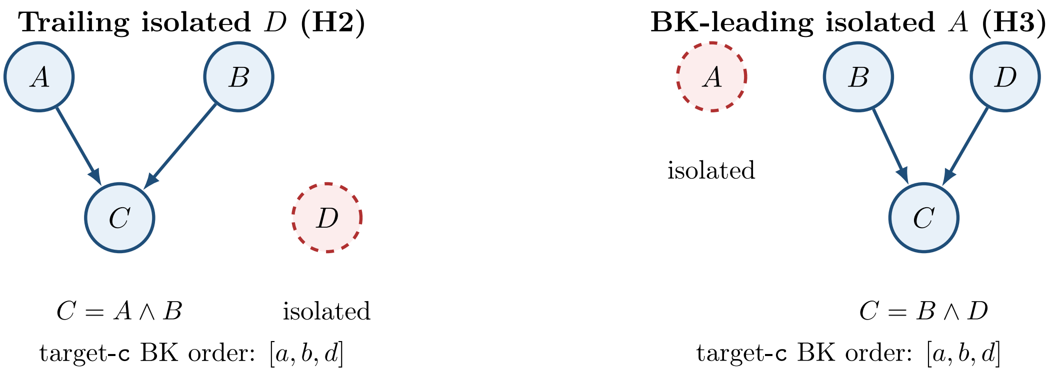

Figure 5: Controlled four-variable extensions. Dashed red nodes are independent isolated variables. Serialisation order induces the stated BK predictor order for target `c`.

Trailing isolated D (H2). This fixture preserves the H0 collider, the H0 root laws $A \sim \text{Bernoulli}(4/5)$ and $B \sim \text{Bernoulli}(7/10)$, and the mechanism $C = A \wedge B$. It adds an independent isolated root $D \sim \text{Bernoulli}(1/2)$ with no edges. Variable order places `d` last, so for target `c` the exact-value BK predictors appear as $[a, b, d]$: the distractor is trailing. The evaluator rule for `c` remains

```
c(A) :- a_val_1(A), b_val_1(A).
```

The variables A, B, and the isolated D are each generated stochastically. When any of them is the learning target, there is no underlying deterministic rule from the remaining observed variables that should be learned. The frozen table used below is again an IID sample with $n = 30$ and seed 42 on this four-variable process.

BK-leading isolated A (H3). This fixture uses independent roots

$$A \sim \text{Bernoulli}(1/2), \qquad B \sim \text{Bernoulli}(4/5), \qquad D \sim \text{Bernoulli}(7/10),$$

with A isolated (no edges) and deterministic child $C = B \wedge D$. Variable order places `a` first, so for target `c` the BK predictors appear as $[a, b, d]$: the distractor is leading. The evaluator rule for `c` is

```
c(A) :- b_val_1(A), d_val_1(A).
```

The isolated variable A and the roots B and D are each generated stochastically. When any of them is the learning target, there is no underlying deterministic rule from the remaining observed variables that should be learned. The frozen table used below is an IID sample with $n = 30$ and seed 42 on this process.

Both extensions have verified Markov factorisation and exact ordinary faithfulness certificates. As with the principal fixture, the generating graph, certificates, and mechanism references are withheld from the learner.

4 Experimental design and comparison logic

This section records the learner configurations used in the probes, how strongly each designed contrast supports attribution to a single change, and the ordered probe map. Detailed outcomes appear in the experimental sequence that follows.

4.1 Learner configurations

Identity	Semantics	Folding	Sel.	Space	Steps	`asm_intro`
`aamas2025`	brave	greedy	mgr	BK	—	`relto`
`ecai2024`	brave	nd	any	all	10	`relto`
`baseline_cautious`	cautious	nd	any	all	10	`relto`
`greedy_cautious`	cautious	greedy	mgr	BK	—	`relto`
`nd_brave_sechk`	brave	nd	any	all	10	`sechk`
`nd_cautious_sechk`	cautious	nd	any	all	10	`sechk`

Here “Steps” is `folding_steps`, which bounds the non-deterministic folding-token budget; it is unused under greedy folding. All six identities also enable the post-learning checked-ASP artefact (`check_ic`). The cautious nd identities additionally pin `post_folding_test_entailment(true)`, which is likewise the effective default under the brave configurations used here.

`aamas2025` and `ecai2024` are the repository presets corresponding to the AAMAS 2025 and ECAI 2024 learning-option bundles used in this work. `baseline_cautious`, `greedy_cautious`, `nd_brave_sechk`, and `nd_cautious_sechk` are experimental identities for controlled comparison. In particular, `greedy_cautious` matches `aamas2025` except for `learning_mode(cautious)`; it is an experimental cautious Greedy preset, not a published AAMAS method.

The H6a configurations match `baseline_cautious` except for `folding_steps`, with $M \in \{1, 2, 5, 10\}$. H7a matches `ecai2024` except `asm_intro(sechk)`. H7b matches `baseline_cautious` except `asm_intro(sechk)`.

4.2 Strength of comparisons

Each probe is read at one of three attribution strengths.

- **Exact paired option ablation.** Fixture, sample, encoding, and the remaining learner options are held fixed; one learning option changes. Probes H4 and H4b change `learning_mode` under otherwise fixed nd / `relto` settings; H5 changes `learning_mode` under the fixed Greedy / `mgr` / BK / `relto` bundle; H6a changes `folding_steps`; H7a and H7b change only `asm_intro`.

- **Controlled data or fixture intervention.** The learner configuration is held fixed (or compared only within one configuration); one designed property of the sample or fixture changes. H1 uses a nested seed-42 prefix $n = 25$ of the principal table under AAMAS; H2 adds a trailing isolated D under AAMAS; H3 uses the BK-leading isolated-A fixture under ECAI (with AAMAS as secondary); H6b varies nested sample size $n \in \{30, 60, 90\}$ at fixed `folding_steps(2)` under `baseline_cautious`.
- **Bundle-level contrast.** Several learner settings change together, so a behavioural difference is attributed to the configuration bundle rather than to one option. H0 compares `aamas2025` with `ecai2024` on the principal table. Where both exist, contrasts between `greedy_cautious` and `baseline_cautious` are likewise bundle-level (Greedy versus nd), as used secondarily in H5.

The strength level states how far a difference among outcomes, deltas, or traces can be attributed to the designed intervention of that probe.

4.3 Probe sequence

Probe	Question	Controlled intervention	What is inspected
H0	What do the Greedy and nd bundles learn for each target on the principal AND collider?	Principal fixture, frozen $n = 30$ seed 42; `aamas2025` and `ecai2024`; targets `a`,`b`,`c`	Exit status, deltas, and traces by target and arm.
H1	How does omitting the rare $(0, 0, 0)$ atom affect AAMAS root learning?	Nested seed-42 prefix $n = 25$ versus $n = 30$; AAMAS; same stream	Root deltas and the presence or absence of opposite-label witness rows.
H2	What does Greedy do with an irrelevant trailing covariate on the child?	Principal AND collider plus isolated D; AAMAS; target `c`	Whether `d` appears in the learned bodies relative to the D-free evaluator rule.
H3	What does nd learning do when an isolated covariate is first in BK?	BD AND collider with BK-leading isolated A; order $[a, b, d]$; ECAI primary on target `c`	First fold, retained BK literals, and the final child theory.
H4	How does switching brave nd to cautious nd affect root targets on the principal table?	Learning mode only, under fixed nd / `relto`; targets `a`,`b` with control `c`	Root exit status and the control child delta.
H4b	How does the same brave-to-cautious switch affect the child repair on the BK-leading fixture?	Learning mode only on H3 target `c`; nd / `relto` fixed	Shared structure versus the contrary-rule close under `alpha_3`.
H5	How does brave-to-cautious behave inside the Greedy bundle?	`learning_mode` only under Greedy / `mgr` / BK / `relto`; 18 paired cells across the completed AAMAS collections	Paired outcomes, deltas, traces, and runtime; secondary contrast to nd cautious where available.
H6a	How does the folding-token ceiling affect cautious-nd search cost on roots?	`folding_steps` $M \in \{1, 2, 5, 10\}$; principal table $n = 30$; `baseline_cautious`	Trace length, runtime, token bands, and control child behaviour.

Probe	Question	Controlled intervention	What is inspected
H6b	How does grounded sample size affect the same cautious-nd search?	Nested $n \in \{30, 60, 90\}$ at fixed $M = 2$; principal fixture; `baseline_cautious`	Search topology and per-band growth with n.
H7a	How do `relto` and `sechk` differ under brave nd?	`asm_intro` only; principal and BK-leading fixtures; seven target pairs	Paired deltas and the assumption-introduction control-flow fork.
H7b	How do `relto` and `sechk` differ under cautious nd?	`asm_intro` only; same two fixtures; `baseline_cautious` versus `nd_cautious_sechk`	Child deltas, root search cost, and whether runs complete within the wall-time limit.

5 Experimental sequence: H0–H7b

5.1 H0: baseline on the principal AND collider

Question and purpose. On the principal fixture $A \to C \leftarrow B$ with $C = A \wedge B$, frozen sample $n = 30$ seed 42, and exact-value encoding, what ABA frameworks do the Greedy bundle (`aamas2025`) and the nd bundle (`ecai2024`) emit when each of `a`, `b`, and `c` is the learning target?

Controlled comparison. The fixture, table, encoding, and target-specific examples were held fixed. The two arms differ as full configuration bundles: `aamas2025` uses brave learning with Greedy / `mgr` / BK folding and `relto`; `ecai2024` uses brave learning with nd / `any` / `all` folding and `relto`. The contrast is therefore bundle-level.

Arm	Target	Outcome	Emphasised result
AAMAS	`c`	`solved`	Assumption-free rule `c(A) :- a_val_1(A), b_val_1(A).`, string-identical to the evaluator mechanism.
AAMAS	`a,b`	`completed_no_solution`	Both-zero co-occurrence candidates cover opposite-label rows; contrary repair dead-ends under `relto`.
ECAI	`c`	`solved`	First-BK under-fold to `a_val_1`, repaired by an assumption attacked when `b_val_0`.
ECAI	`a,b`	`solved`	Brave per-row assumption construction on the ambiguous predictor cell; A (resp. B) remains underdetermined by the other observed variables.

Child target `c`

Emphasised observation. Under AAMAS, every positive example for `c` has $(A, B) = (1, 1)$. Greedy folding with BK space therefore builds the single maximal co-occurrence body $\{\texttt{a\_val\_1}, \texttt{b\_val\_1}\}$. After mgr deduplication the emitted delta is

```
c(A) :- a_val_1(A), b_val_1(A).
```

Brave entailment succeeds immediately: no negative example shares that body, so no assumption is introduced. The string coincides with the evaluator mechanism reference for C.

Under ECAI, nd folding takes one BK predicate at a time. From an early positive, the first fold replaces the example identifier by `a_val_1`, the first predictor in BK order, yielding the unary rule `c(A) :- a_val_1(A)`. That body also holds of every negative with $A = 1, B = 0$. Assumption introduction produces

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

Trace-supported explanation. The ECAI theory is sample-adequate under brave checking: c is derived when $a = 1$, unless the assumption is attacked when $b = 0$. Across the target rule and the contrary it cites both parents of C. Its ABA shape is nevertheless different from the AAMAS conjunction: the first fold latched onto the leading BK predictor, and the second parent enters only through contrary learning.

Root targets `a` and `b`

Emphasised observation. For target `a`, positives fall into two predictor cells, $(B, C) = (1, 1)$ and $(B, C) = (0, 0)$. The cell $(B, C) = (0, 0)$ also contains negatives: row 9 is positive with joint values $(1, 0, 0)$, while rows 26 and 30 are negative with $(0, 0, 0)$, yet all share the BK literals `b_val_0` and `c_val_0`. Target `b` is the symmetric case with the roles of A and B swapped.

Under AAMAS, greedy folding proposes the two Horn candidates

```
a(A) :- b_val_1(A), c_val_1(A).
a(A) :- b_val_0(A), c_val_0(A).
```

The both-one rule is accepted. The both-zero rule covers the negative rows 26 and 30, so brave entailment fails. The learner introduces an assumption on that body; contrary rote learning then cites those negatives, and greedy folding of the contrary reconstructs the same both-zero body. Under `asm_intro(relto)` a further relative assumption cannot be added for that body, the engine reports KO, and the cell ends as `completed_no_solution` with no emitted delta. Target `b` mirrors this path.

Under ECAI, nd folding again proceeds one predicate at a time. The learner first builds a safe block for the both-one positives,

```
a(A) :- alpha_1(A), b_val_1(A).
c_alpha_1(A) :- c_val_0(A).
```

That block covers positives with $(B, C) = (1, 1)$ and excludes negatives with $B = 1$ and $C = 0$. The both-zero positives remain uncovered, so a second ordinary target rule is still required.

Folding an early both-zero positive (row 8) produces the candidate `a(A) :- b_val_0(A)`. Brave entailment fails on that candidate. The same BK literals also hold of the negative examples in rows 26 and 30. The learner therefore introduces an assumption α_2 on the body, giving `a(A) :- alpha_2(A), b_val_0(A)`. Contrary rote learning cites rows 26 and 30 as grounds for `c_alpha_2`.

Folding that contrary first tries `b_val_0`. Under `relto` this returns KO against the already present α_2. The next fold tries `c_val_0`, yielding `c_alpha_2(A) :- c_val_0(A)`. Even that unary contrary remains too strong for the joint example set. The learner therefore introduces a further assumption α_3 on the contrary. Rote learning of `c_alpha_3` then cites the both-zero positives, including row 9. Folding that contrary re-uses α_2 together with `b_val_0`. The trace records this close as `found:`

`alpha_2/1 ...  OK`. The final delta for target `a` is

```
a(A) :- alpha_1(A), b_val_1(A).
a(A) :- alpha_2(A), b_val_0(A).
c_alpha_1(A) :- c_val_0(A).
c_alpha_2(A) :- alpha_3(A), c_val_0(A).
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
assumption(alpha_1(A)).
assumption(alpha_2(A)).
assumption(alpha_3(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
contrary(alpha_2(A),c_alpha_2(A)) :- assumption(alpha_2(A)).
contrary(alpha_3(A),c_alpha_3(A)) :- assumption(alpha_3(A)).
```

Target `b` is the mirror obtained by swapping the roles of A and B.

Trace-supported explanation. The decisive geometry is the predictor cell $(B, C) = (0, 0)$. Row 9 is a positive example for target `a`. Its joint values are $(A, B, C) = (1, 0, 0)$. Row 26 is a negative example. Its joint values are $(0, 0, 0)$. Both rows carry the BK facts `b_val_0` and `c_val_0`. Any Horn rule that uses only those predictors therefore treats the two rows alike. It would derive `a` at both identifiers, or at neither.

The emitted α_2/α_3 nest works because assumptions are grounded per row identifier. At a fixed row i, the relevant instances are $\alpha_2(i)$ and $\alpha_3(i)$, with contraries `c_alpha_2`(i) and `c_alpha_3`(i). The local rules are

```
a(i) :- alpha_2(i), b_val_0(i).
c_alpha_2(i) :- alpha_3(i), c_val_0(i).
c_alpha_3(i) :- alpha_2(i), b_val_0(i).
```

On every both-zero row, both `b_val_0(i)` and `c_val_0(i)` hold. Accepting $\alpha_2(i)$ therefore derives `c_alpha_3`(i) and attacks $\alpha_3(i)$. Accepting $\alpha_3(i)$ derives `c_alpha_2`(i) and attacks $\alpha_2(i)$. The two assumptions cannot both be accepted at the same row.

This leaves two locally stable choices at each both-zero identifier. If $\alpha_2(i)$ is accepted and $\alpha_3(i)$ is rejected, then `a(i)` is derived. If $\alpha_3(i)$ is accepted and $\alpha_2(i)$ is rejected, then `a(i)` is blocked.

Brave learning asks for one witnessing stable extension Δ of the whole grounded framework. That extension must cover every positive example and exclude every negative example at once. Such a witness exists. On row 9 it accepts $\alpha_2(9)$ and rejects $\alpha_3(9)$. The rule `a(9) :- alpha_2(9), b_val_0(9)` then derives the positive example $(1, 0, 0)$. On row 26 it accepts $\alpha_3(26)$ and rejects $\alpha_2(26)$. The same BK facts are present, but without $\alpha_2(26)$ the rule for `a` does not fire, so the negative example $(0, 0, 0)$ is excluded. The two rows are therefore separated by their ground assumption atoms, even though their BK predictors are identical.

Row	Joint (A, B, C)	Choice in Δ	Effect
9	$(1, 0, 0)$ (positive)	$\alpha_2(9)$ in, $\alpha_3(9)$ out	`a(9)` derived
26	$(0, 0, 0)$ (negative)	$\alpha_2(26)$ out, $\alpha_3(26)$ in	`a(26)` blocked

Other stable extensions may reverse those local choices. Brave entailment requires only that at least one joint witness exists. The framework therefore passes brave checking on the finite sample.

For this mechanism, target `a` should have no learned solution: A is generated stochastically, and there is no underlying deterministic rule from the other observed variables. A solution was nevertheless found. Brave entailment permits one stable extension to assign opposite statuses to the ground assumptions $\alpha_2(i)$ and $\alpha_3(i)$ at different row identifiers. That per-row split covers the positive example $(1, 0, 0)$ and excludes the negative example $(0, 0, 0)$ even though the two rows share the same BK predictors. The ECAI `solved` outcome records that brave covering. It does not record recovery of a deterministic rule for A.

Evidential boundary. H0 is the establishing bundle contrast on one complete-support table. Procedural `solved` covers both the AAMAS mechanism-string match for `c` and the ECAI assumption-rich theories for `c` and the roots. Attribution of the arm difference is to the full AAMAS versus ECAI option bundle.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/learning_analysis.md`

5.2 H1: nested prefix omitting the rare $(0, 0, 0)$ atom

Question and purpose. Under the fixed AAMAS bundle on the principal AND collider, what happens to the root targets when the seed-42 table is replaced by its nested prefix of length 25, which omits the only sample atoms $(0, 0, 0)$?

Controlled comparison. Everything except sample size is held fixed. H0 uses the first 30 rows of the seed-42 stream. H1 uses the exact first 25 rows of that same stream. The omitted rows are CSV rows 26 and 30, the only H0 occurrences of $(0, 0, 0)$. The population still assigns mass $P^*(0, 0, 0) = 3/50$ to that atom.

Target	H0 ($n = 30$)	H1 ($n = 25$)
`a`,`b`	`completed_no_solution`	`solved` with both-one and both-zero Horn rules
`c`	`solved` with the evaluator AND string	`solved` with the same AND string

Emphasised observation. The H1 root delta for target `a` is

```
a(A) :- b_val_1(A), c_val_1(A).
a(A) :- b_val_0(A), c_val_0(A).
```

Target `b` emits the mirror pair. The both-zero rules are false in the population. They are sample-adequate on H1 only because the prefix contains no negative example in the predictor cell $(B, C) = (0, 0)$ (resp. $(A, C) = (0, 0)$).

Trace-supported explanation. H0 already showed that, under AAMAS, the both-zero body is proposed for root `a` and is rejected when rows 26 and 30 are present as negatives in that cell. H1 removes exactly those rows. The same body then faces only positives in that cell, so brave entailment accepts it and the run ends as `solved`.

For these stochastic roots, the intended reading is that no deterministic rule from the other observed variables should be learned. The H0 outcome `completed_no_solution` matches that reading. The H1 outcome `solved` is a spurious finite-sample covering created by omitting the rare blocking atom. Child `c` is unchanged and serves only as a control.

Evidential boundary. H1 is a nested-prefix support ablation under fixed AAMAS settings.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h1_support_ablation.md`

5.3 H2: trailing isolated covariate under Greedy

Question and purpose. On the principal AND collider extended by an independent isolated root D, does AAMAS learning of the deterministic child `c` retain irrelevant `d` literals, or does it return the evaluator D-free rule `c(A) :- a_val_1(A), b_val_1(A).`?

Controlled comparison. The H0 collider, root laws, and mechanism $C = A \wedge B$ are preserved. An isolated root $D \sim \text{Bernoulli}(1/2)$ is added with no edges. Variable order places `d` last, so for target `c` the BK predictors are $[a, b, d]$. The run uses AAMAS on the frozen $n = 30$ seed-42 sample. Among the eighteen positive `c` rows, both $D = 1$ and $D = 0$ occur nine times.

Emphasised observation. AAMAS solves target `c` and emits

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_1(A), b_val_1(A), d_val_0(A).
```

Irrelevant D is retained. The two rules are sample-adequate and jointly cover the observed values of D, but they are non-minimal relative to the evaluator mechanism

```
c(A) :- a_val_1(A), b_val_1(A).
```

On the three-variable H0 table without D, the same AAMAS bundle emitted exactly that D-free string.

Trace-supported explanation. Under greedy folding with `folding_space(bk)`, each positive example is folded into a maximal co-occurrence body from the BK literals that hold of that row. A positive with $D = 1$ therefore keeps `d_val_1`. A positive with $D = 0$ keeps `d_val_0`. The trace shows this explicitly: after `a_val_1` and `b_val_1`, the next fold attaches the matching `d_val_*` literal. Mgr selection then keeps the resulting bodies. There is no later step that merges the complementary pair by deleting the `d` literal. The failure mode under test is therefore distractor retention on a solved child, not failure to solve `c`.

Evidential boundary. H2 is a controlled fixture intervention under fixed AAMAS settings. The primary signal is the balanced $n = 30$ distractor split on target `c`.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h2_irrelevant_covariate.md`

5.4 H3: BK-leading isolated covariate under nd

Question and purpose. On a BD AND collider with an isolated root placed first in BK order, does ECAI learning of the deterministic child `c` select that isolated variable in the first fold and retain it in the final theory, or does it return the evaluator rule `c(A) :- b_val_1(A), d_val_1(A).`?

Controlled comparison. The fixture is $B \to C \leftarrow D$ with $C = B \wedge D$ and an independent isolated root $A \sim$ Bernoulli(1/2). Variable order places `a` first, so for target `c` the BK predictors are $[a, b, d]$. The primary run is ECAI brave nd on the frozen $n = 30$ seed-42 sample. An AAMAS run on the same table supplies a bundle-level contrast. Among the fifteen positive `c` rows, every row has $(B, D) = (1, 1)$, while both values of A occur.

Emphasised observation. The ECAI trace opens with the first fold

```
folding result: c(A) <- [a_val_1(A)]
```

Assumption introduction follows immediately. The full emitted delta for target `c` is:

```
c(A) :- alpha_1(A), a_val_1(A).
c(A) :- alpha_2(A), a_val_0(A).
c_alpha_1(A) :- b_val_0(A).
c_alpha_1(A) :- alpha_3(A), b_val_1(A).
c_alpha_2(A) :- d_val_0(A).
c_alpha_2(A) :- b_val_0(A).
c_alpha_3(A) :- alpha_1(A), a_val_1(A).
assumption(alpha_1(A)).
assumption(alpha_2(A)).
assumption(alpha_3(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
contrary(alpha_2(A),c_alpha_2(A)) :- assumption(alpha_2(A)).
contrary(alpha_3(A),c_alpha_3(A)) :- assumption(alpha_3(A)).
```

The two ordinary target rules are gated only on the isolated variable A. Mechanism parents enter through the contrary layer: `c_alpha_1` and `c_alpha_2` cite b and d, with a further nested assumption α_3 under the a=1 branch. The theory is therefore organised around A, with B and D used to repair that scaffolding. It does not match the evaluator BD AND `c(A) :- b_val_1(A), d_val_1(A).`.

On the same sample, AAMAS also solves `c` while retaining `a`:

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).
c(A) :- a_val_0(A), b_val_1(A), d_val_1(A).
```

Trace-supported explanation. Under nd folding with `any` selection, the first admissible BK predicate becomes the first generalising fold. Here that predicate is an `a`-value literal, because `a` leads the serialised BK. The unary fold over-generalises, so the learner introduces α_1 (then α_2) and learns contraries around those A-gated rules. Later folds place b and d in `c_alpha_1` and `c_alpha_2`, but the ordinary target rules keep the initial a-gates. The AAMAS contrast reaches distractor retention by a different route: greedy maximal co-occurrence keeps every matching literal, including leading `a`, and splits the AND across the two values of A.

In both cases the child is solved, and in both cases the isolated leading variable is retained. The failure mode is distractor inclusion driven by BK order, not failure to solve `c`.

Contrary-structure divergence under the two A gates. The ECAI delta is asymmetric under the two leading-a gates. That asymmetry is part of the H3 theory and deserves a separate reading.

Under a=1 (α_1), the contrary of α_1 is *nested*:

```
c_alpha_1(A) :- b_val_0(A).
```

```
c_alpha_1(A) :- alpha_3(A), b_val_1(A).
c_alpha_3(A) :- alpha_1(A), a_val_1(A).
```

Under a=0 (α_2), the contrary of α_2 is *flat* and cites the mechanism parents directly:

```
c_alpha_2(A) :- d_val_0(A).
c_alpha_2(A) :- b_val_0(A).
```

So one gate is repaired by a further assumption α_3, while the other is repaired by ordinary b/d literals alone.

The divergence is a search-order effect under nd folding with `any` selection, b-before-d BK order after leading `a`, cover-and-repair, and `asm_intro(relto)`. It is not a difference in the underlying mechanism of C, and it is not caused by the brave/cautious learning-mode choice — the same α_1-nested / α_2-flat skeleton appears under both H3 brave and H4b cautious. Concretely:

1. Contrary repair for `c_alpha_1` occurs first, because the first positive example has a=1.
2. Under that gate, the first contrary seed has $(b, d) = (0, 1)$, so search commits to `b_val_0`. The remaining contrary residuals have $(b, d) = (1, 0)$.
3. Folding those residuals with `b_val_1` over-generalises. Instead of backtracking to a flat cover $\{$`b_val_0`, `d_val_0`$\}$, cover-and-repair introduces α_3 on the `b_val_1` body.
4. When search later repairs `c_alpha_2` under a=0, a fold that would reuse `b_val_1` meets the already declared α_3 under `relto` and is rejected. Search then falls through to `d_val_0` and `b_val_0`, yielding the flat contrary.

Thus the nested α_1 side and the flat α_2 side are two stages of one sequential repair path. The second gate inherits the assumption minted while repairing the first.

Evidential boundary. H3 is a controlled fixture intervention. The primary signal is ECAI first-fold retention of BK-leading isolated A on target `c`. The α_1-nested versus α_2-flat contrary split is a secondary procedural feature of that same theory.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h3_bk_leading_distractor.md`

5.5 H4: brave versus cautious under nd

5.5.1 H4: roots on the H0 table

Question and purpose. On the principal AND-collider table of H0, does changing `learning_mode(brave)` to `learning_mode(cautious)` alter the outcomes for root targets `a` and `b` that, under `ecai2024`, relied on a residual per-row assumption construction separating rows 9 and 26, and what happens to control target `c`?

Controlled comparison. The H0 fixture, frozen $n = 30$ seed-42 table, target-specific examples, and BK were held fixed. The two learner configurations compared are `ecai2024` and `baseline_cautious`. Both use nd folding with `folding_selection(any)`, `folding_space(all)`, `asm_intro(relto)`, and `folding_steps(10)`. They differ in learning mode: `ecai2024` sets `learning_mode(brave)`; `baseline_cautious` sets `learning_mode(cautious)`. The locked `ecai2024` outputs are the brave comparator.

Target	ecai2024	baseline_cautious
a	solved	completed_no_solution
b	solved	completed_no_solution
c	solved	solved, byte-identical delta

Emphasised observation. For target a, the ecai2024 and baseline_cautious traces coincide through the early $\alpha_1, \alpha_2, \alpha_3$ construction from H0. They diverge at one assumption-introduction step. After

```
folding result: c_alpha_3(A) <- [b_val_0(A)]
```

the learner proposes to re-use the already declared assumption α_2 in that body:

```
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

Under ecai2024 the proposal is accepted; under baseline_cautious it is rejected:

```
% ecai2024:
 found: alpha_2/1 ... OK, assumption introduction result:
   c_alpha_3(A) <- [alpha_2(A),b_val_0(A)]

% baseline_cautious:
 found: alpha_2/1 ... KO, cannot introduce an assumption for [b_val_0(A)]
```

Under ecai2024, that OK closes the both-zero nest from H0 and the learner writes a solution. Under baseline_cautious, the KO is followed by further failed folds, exhaustion of folding_steps(10), and completed_no_solution with no emitted delta. Target b exhibits the same divergence with the roles of A and B swapped.

Under baseline_cautious, control target c remains solved. Its delta is byte-identical to the locked ecai2024 delta for c:

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
```

Thus learning_mode(cautious) still admits an assumption-bearing theory for the child on this table, while rejecting the residual α_2 reuse for the roots.

Trace-supported explanation. Under both configurations the search builds toward the same both-zero nest:

```
a(A) :- alpha_2(A), b_val_0(A).
c_alpha_2(A) :- alpha_3(A), c_val_0(A).
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

Write $\mathcal{F}_{\text{nest}}$ for a framework containing this nest. The decisive conflict is observational. Row 9 contributes $a(9) \in E^+$ with joint values $(A, B, C) = (1, 0, 0)$. Row 26 contributes $a(26) \in E^-$ with joint values $(0, 0, 0)$. Relative to BK the two rows are indistinguishable: both carry b_val_0 and c_val_0. No Horn body over those predictors alone can accept one row and reject the other.

The nest does not separate the rows by BK. It separates them by ground assumption membership.

At any both-zero row identifier i, the assumptions $\alpha_2(i)$ and $\alpha_3(i)$ attack each other through the contrary layer, so a stable extension can take only one of the two local choices:

Choice at row i	Acceptance of $a(i)$	Derived contrary
$\alpha_2(i) \in \Delta$, $\alpha_3(i) \notin \Delta$	$\mathcal{F}_{\text{nest}} \models_\Delta a(i)$	`c_alpha_3`(i)
$\alpha_2(i) \notin \Delta$, $\alpha_3(i) \in \Delta$	$\mathcal{F}_{\text{nest}} \not\models_\Delta a(i)$	`c_alpha_2`(i)

Brave entailment. Under `learning_mode(brave)`, a learned framework $\mathcal{F}'$ is accepted when there exists one witnessing stable extension covering the examples:

$$\exists \Delta \in \mathrm{Stab}(\mathcal{F}') \left[\forall e \in E^+,\ \mathcal{F}' \models_\Delta e \ \wedge\ \forall e \in E^-,\ \mathcal{F}' \not\models_\Delta e\right].$$

For $\mathcal{F}_{\text{nest}}$ such a witness exists. On row 9 it chooses $\alpha_2(9)$ in and $\alpha_3(9)$ out, so

```
a(9) :- alpha_2(9), b_val_0(9)
```

yields $\mathcal{F}_{\text{nest}} \models_\Delta a(9)$. On row 26 it chooses $\alpha_2(26)$ out and $\alpha_3(26)$ in, so $\mathcal{F}_{\text{nest}} \not\models_\Delta a(26)$. The closing rule

```
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

is part of making those local choices coherent: whenever $\alpha_2(i)$ is accepted, `c_alpha_3`(i) is derived and attacks $\alpha_3(i)$.

Thus $\mathcal{F}_{\text{nest}}$ passes brave entailment because coverage may depend on different ground assumption choices at different row identifiers inside a single witness. The assumption-introduction check that proposes re-using α_2 on `c_alpha_3(A) <- [b_val_0(A)]` therefore returns OK, and `ecai2024` finishes with `solved`.

Failure under cautious entailment. Under `learning_mode(cautious)`, the same framework must satisfy the stronger condition

$$\forall e \in E^+,\ \mathcal{F}' \models_{\text{caut}} e, \qquad \forall e \in E^-,\ \mathcal{F}' \not\models_{\text{caut}} e,$$

where $\mathcal{F}' \models_{\text{caut}} s$ means that s is accepted with respect to every stable extension of $\mathcal{F}'$.

Apply that test to $\mathcal{F}_{\text{nest}}$ at row 9. The choice that covers the positive is $\alpha_2(9) \in \Delta$. The opposite choice, $\alpha_3(9) \in \Delta$ and $\alpha_2(9) \notin \Delta$, also belongs to $\mathrm{Stab}(\mathcal{F}_{\text{nest}})$. In any extension Δ' that takes the opposite choice, $\mathcal{F}_{\text{nest}} \not\models_{\Delta'} a(9)$. Therefore $a(9)$ fails $\mathcal{F}_{\text{nest}} \models_{\text{caut}} a(9)$: it is not a cautious consequence. The brave cover of the positive exists, but it is only an existential cover. Cautious entailment requires universal cover, and this nest cannot provide it on an underdetermined BK cell.

That is why, under `baseline_cautious`, the same proposal returns

```
found: alpha_2/1 ... KO, cannot introduce an assumption for [b_val_0(A)]
```

The learner never accepts the clause that would close the brave construction. Search then exhausts `folding_steps(10)` and returns `completed_no_solution`.

In short: `ecai2024` succeeds because brave entailment needs only some stable extension that assigns opposite α_2/α_3 statuses to rows 9 and 26; `baseline_cautious` fails because $a(9)$ is not accepted in every stable extension of that same nest, so the residual α_2 reuse cannot pass the cautious entailment check.

The control on target `c` is consistent with these semantics. Its contrary folds to the ordinary body `b_val_0`, which does not require opposite assumption membership across rows that share the same

BK literals. The cautious entailment checks therefore succeed, and the emitted delta matches the `ecai2024` delta for `c`.

Evidential boundary. H4 compares learner configurations `ecai2024` and `baseline_cautious` on the locked H0 table, with matched nd folding options, matched `asm_intro(relto)`, and `folding_steps(10)`. Root `completed_no_solution` under `baseline_cautious` is the absence of an accepted covering theory in that setting.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h4_cautious_vs_brave.md`

5.5.2 H4b: child c on the H3 table

Question and purpose. On the H3 BD AND table with BK-leading isolated A, under matched nd and `relto` settings, does changing `learning_mode(brave)` to `learning_mode(cautious)` alter how the contrary of α_3 is closed when learning target `c`, and what of the H3 theory remains unchanged?

Controlled comparison. The H3 fixture `m13_bucket3_binary_bd_and_lead_a`, frozen $n = 30$ seed-42 sample, target `c`, and BK order $[a, b, d]$ were held fixed. The two learner configurations compared are `ecai2024` and `baseline_cautious`. Both use nd folding with `folding_selection(any)`, `folding_space(all)`, `asm_intro(relto)`, and `folding_steps(10)`. They differ in learning mode: `ecai2024` sets `learning_mode(brave)`; `baseline_cautious` sets `learning_mode(cautious)`. The locked H3 `ecai2024` cell for target `c` is the brave comparator. This probe concerns nd folding under cautious learning; it is distinct from the later Greedy probe H5.

Object	`ecai2024`	`baseline_cautious`
Outcome for `c`	`solved`	`solved`
Target gates	$\alpha_1, a{=}1$ / $\alpha_2, a{=}0$	same
`c_alpha_1`	$b{=}0$; nest $\alpha_3, b{=}1$	same
`c_alpha_2`	$d{=}0$; $b{=}0$	same
`c_alpha_3`	$\alpha_1, a{=}1$	$d{=}1$

Emphasised observation. Both configurations solve target `c`. Through the introduction of α_3, their theories coincide with the H3 skeleton already recorded: ordinary target rules gated on leading a, a nested contrary under $a{=}1$, and a flat b/d contrary under $a{=}0$. Exactly one closing rule differs. Under `ecai2024`:

```
c_alpha_3(A) :- alpha_1(A), a_val_1(A).
```

Under `baseline_cautious`:

```
c_alpha_3(A) :- d_val_1(A).
```

The traces diverge at the assumption-introduction check on that close. After

```
folding result: c_alpha_3(A) <- [a_val_1(A)]
```

`ecai2024` re-uses α_1 and stops:

```
found: alpha_1/1 ... OK, assumption introduction result:
  c_alpha_3(A) <- [alpha_1(A),a_val_1(A)]
```

`baseline_cautious` rejects that reuse, rejects a subsequent attempt to re-use α_3 on `b_val_1`, and then accepts an ordinary `d_val_1` fold:

```
found: alpha_1/1 ... KO, cannot introduce an assumption for [a_val_1(A)]
folding result: c_alpha_3(A) <- [b_val_1(A)]
found: alpha_3/1 ... KO, cannot introduce an assumption for [b_val_1(A)]
folding result: c_alpha_3(A) <- [d_val_1(A)]
```

The full `baseline_cautious` delta is therefore the shared H3 skeleton with that single close replaced:

```
c(A) :- alpha_1(A), a_val_1(A).
c(A) :- alpha_2(A), a_val_0(A).
c_alpha_1(A) :- b_val_0(A).
c_alpha_1(A) :- alpha_3(A), b_val_1(A).
c_alpha_2(A) :- d_val_0(A).
c_alpha_2(A) :- b_val_0(A).
c_alpha_3(A) :- d_val_1(A).
assumption(alpha_1(A)).
assumption(alpha_2(A)).
assumption(alpha_3(A)).
contrary(alpha_1(A),c_alpha_1(A)) :- assumption(alpha_1(A)).
contrary(alpha_2(A),c_alpha_2(A)) :- assumption(alpha_2(A)).
contrary(alpha_3(A),c_alpha_3(A)) :- assumption(alpha_3(A)).
```

Trace-supported explanation. The shared prefix is the H3 theory under leading a. As recorded in H3, that prefix is produced by nd folding with `any` selection, b-before-d BK order, cover-and-repair, and `asm_intro(relto)`. It is reproduced under both learning modes through α_3 introduction. The α_1-nested versus α_2-flat contrary split is therefore already present before the learning-mode contrast takes effect.

The learning-mode contrast is confined to the attack on α_3. Under `ecai2024`, the learner closes that attack by residual reuse of α_1:

```
c_alpha_3(A) :- alpha_1(A), a_val_1(A).
```

This clause couples the contrary of α_3 to the same assumption that gates the ordinary target rule under a=1. Brave entailment asks only for some witnessing stable extension of the candidate framework that covers E^+ and excludes E^-. That residual reuse passes the brave assumption-introduction check, so `ecai2024` accepts it and never folds a d-literal on `c_alpha_3`.

Under `baseline_cautious`, the entailment condition is universal: every positive must be a cautious consequence, and every negative must be excluded as a cautious consequence. The residual reuse of α_1 on `c_alpha_3` fails that test, exactly as the residual reuse of α_2 failed under H4: acceptance of the proposed extension depends on assumption choices that need not hold in every stable extension. The introduction check therefore returns KO. Unlike H4 on the roots, search here still has an ordinary BK alternative. Folding the contrary examples of α_3 with `d_val_1` satisfies the cautious entailment condition on $\langle E^+, E^-\rangle$, so the learner accepts

```
c_alpha_3(A) :- d_val_1(A).
```

and finishes with `solved`.

Relative to H4, the same learning-mode change therefore has the same local effect—rejection of a residual assumption reuse—but a different global outcome. On the H0 roots, no accepted covering theory remained within `folding_steps(10)`. On this H3 child cell, an ordinary parent literal closes the nest, while the leading-a gates and the asymmetric contrary skeleton from H3 remain in place.

Evidential boundary. H4b compares `ecai2024` and `baseline_cautious` on the locked H3 target-`c` cell, with matched nd folding options and `asm_intro(relto)`. The emitted theories retain BK-leading isolated A in the ordinary target rules. The learning-mode effect isolated here is the body of `c_alpha_3`.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h4b_cautious_split_under_a.md`

5.6 H5: Greedy under cautious versus brave learning

Question and purpose. Under otherwise fixed Greedy options, does changing `learning_mode(brave)` to `learning_mode(cautious)` alter the accepted or rejected theories on the completed Greedy collections from H0–H3?

Controlled comparison. The locked `aamas2025` collections for H0 ($n = 30$), H1 ($n = 25$), H2 ($n = 30$), the extreme H2 edge ($n = 2$), and H3 ($n = 30$) were re-run under the experimental configuration `greedy_cautious`. That configuration matches `aamas2025` on Greedy folding, `folding_selection(mgr)`, `folding_space(bk)`, `asm_intro(relto)`, and `check_ic`, and differs only by setting `learning_mode(cautious)`. This gives eighteen target-wise pairs on shared tables, BK, and examples. Configuration `greedy_cautious` is an experimental preset for this contrast; it is not a published AAMAS method.

Emphasised observation. On all eighteen pairs, `greedy_cautious` matches `aamas2025` on every outcome, and every solved delta is byte-identical. The only systematic difference is runtime: cautious runs take approximately 1.6–2.3 times as long as the corresponding brave runs (mean ratio about 1.84), while remaining below one second.

Trace-supported explanation. The learning-mode change does not alter the accepted or rejected Greedy theories on these cells. The runtime increase comes from the entailment checks performed along the shared search path. Brave checks ask whether some answer set satisfies the joint example constraints. Cautious checks ask Clingo for cautious consequences (`-enum-mode=cautious`) — atoms true in every answer set — and then test E^+ and E^- against that intersection. The same sequence of checks is therefore more expensive under cautious learning.

Evidential boundary. H5 is restricted to these eighteen Greedy pairs under matched non-mode options.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h5_greedy_cautious.md`

5.7 H6: explain cautious-nd search cost

5.7.1 H6a: folding-token ceiling

Question and purpose. On the H0 table under `baseline_cautious`, how do root no-solution trace length and runtime scale with the folding-token ceiling $M =$ `folding_steps`?

Controlled comparison. The H0 fixture, frozen $n = 30$ seed-42 sample, target-specific examples, BK, `learning_mode(cautious)`, nd folding, `folding_selection(any)`, `folding_space(all)`, and `asm_intro(relto)` were held fixed. The single varied option was `folding_steps(M)` for $M \in \{1, 2, 5, 10\}$. Configurations `baseline_cautious_steps1`, `baseline_cautious_steps2`, and `baseline_cautious_steps5` match `baseline_cautious` except for M. The $M = 10$ collection reuses the locked H4 run under `baseline_cautious`. Roots `a` and `b` are the intended no-solution targets (stochastic; no desired deterministic rule from the other observed variables). Child `c` is the control.

M	Target	Outcome	Runtime (s)	Trace lines	New assumptions
1	a	`completed_no_solution`	6.638	549	9
2	a	`completed_no_solution`	13.117	1047	18
5	a	`completed_no_solution`	32.775	2541	45
10	a	`completed_no_solution`	69.952	5031	90

Target b mirrors these figures. Control `c` solves at every M inside `tokens(1)`: 134-line trace, about 1.3 s, and a byte-identical delta across all four ceilings.

Emphasised observation. For roots `a` and `b` the outcome is `completed_no_solution` at every tested M: no covering theory is accepted. Each run is a stack of M failed bands. The $M = 1$ trace has 549 lines and nine events `generating NEW assumption:  alpha_k(A)`. Each extra allowance adds 498 lines and nine further new assumptions. At $M = 10$ the trace prints nine restarts

```
* Increasing folding tokens to: 2
...
* Increasing folding tokens to: 10
```

and ends with `* No solution found!`. The counts fit

$$\text{lines}(M) = 549 + 498(M - 1) = 51 + 498\,M, \qquad \#\{\text{new assumptions}\}(M) = 9M.$$

Runtime grows in the same way. Raising M multiplies failed bands; it does not produce a root solution. The long H4 root traces are the $M = 10$ endpoint of this ladder.

Trace-supported explanation. A failed band is one complete cautious search at fixed token allowance T that exhausts without writing a delta. On these roots the H4 residual assumption reuse already returns KO inside the first allowance, so that first band fails.

Under nd folding, `folding_steps(M)` is the ceiling on how many such attempts may run. Search starts at `tokens(1)`. When the attempt at allowance T fails completely and $T + 1 \leq M$, the controller prints `* Increasing folding tokens to:  \(T{+}1\)` and restarts the same learning problem on the same rules and the same E^+, E^-. The target still admits no cautiously acceptable cover, so each restart fails in the same way. M allowances therefore contribute M copies of the same failed band, which is why trace length and new-assumption counts are linear in M.

Control `c` solves under `tokens(1)`, never enters the restart ladder, and is therefore invariant in M.

Evidential boundary. H6a is a single-option ablation of `folding_steps` under `baseline_cautious` on the locked H0 $n = 30$ table, for these intended no-solution roots over $M \in \{1, 2, 5, 10\}$.

5.7.2 H6b: nested sample size

Question and purpose. H6b asked how cautious-nd trace size changes when the folding-token topology is held fixed but more grounded IID rows are supplied.

Controlled comparison. The H0 population, seed 42, `baseline_cautious` bundle, and $M = 2$ were fixed. Nested prefixes used $n \in \{30, 60, 90\}$.

n	a lines	a runtime (s)	NEW	KO	c lines
30	1047	13.117	18	32	134
60	1641	22.169	18	32	200
90	2241	32.291	18	32	266

All four H0 population atoms are already present at $n = 30$. Their multiplicities, ordered as $(0, 0, 0), (0, 1, 0), (1, 0, 0), (1, 1, 1)$, are:

$$\begin{aligned} n = 30 &: (2, 4, 7, 17), \\ n = 60 &: (4, 10, 12, 34), \\ n = 90 &: (5, 17, 18, 50). \end{aligned}$$

Emphasised observation. The root search topology stays fixed at 18 new assumptions and 32 KOs, while target-a trace length increases by 594 and 600 lines for each additional 30 rows. Target b mirrors this. Target c retains the same delta but its trace grows by 66 lines per 30 rows.

Trace-supported explanation. Additional rows create more grounded example identifiers, rote rules, entailment checks, subsumption work, and contrary batches inside each fixed search band. They do not introduce a new supported value pattern in this comparison. The approximately linear line growth therefore comes from per-row grounding and bookkeeping while the procedural topology remains unchanged.

Evidential boundary. H6b is a nested-n experiment, not a direct deduplication ablation. Multiplicities still carry empirical frequency information, and removing duplicate rows can alter row order and learner behaviour. The result is local to $n = 30, 60, 90$, not a universal complexity law.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h6_folding_and_n_ablation.md`

5.8 H7: change assumption introduction from `relto` to `sechk`

5.8.1 H7a: brave nd

Question and purpose. H7a asked whether the learned rules and trace path change when the assumption-introduction option is changed inside the brave nd bundle. It also asked whether any difference follows a consistent relationship to serialized BK order and causal role.

Controlled comparison. Within each pair, H0 or H3 fixture, $n = 30$ sample, target, examples, BK, brave semantics, nd folding, `any` selection, `all` space, folding ceiling, and other settings were fixed. The locked `ecai2024/relto` collection was compared with experimental `nd_brave_sechk/sechk`. There were seven target-wise pairs.

The target roles must be kept explicit:

Cell	Causal role	Evaluative use
H0/H3 c	deterministic child	desirable mechanism target
H0 a	parent of c, root as target	root diagnostic
H3 b	parent of c, root as target	role-peer of H0 a
H3 a	isolated root	leading-distractor diagnostic
H0 b, H3 d	other parent roots	supporting mirrors

Emphasised observation. Both arms solve all seven cells, but every paired `delta.aba` differs:

Cell	BK order	Outcome	`sechk` variables	Additional `relto` variables
H0 a	b, c	solved / solved	b	c
H0 b	a, c	solved / solved	a	c
H0 c	a, b	solved / solved	a	b
H3 a	b, c, d	solved / solved	b	c, d
H3 b	a, c, d	solved / solved	a	c, d
H3 c	a, b, d	solved / solved	a	b, d
H3 d	a, b, c	solved / solved	a	b, c

For H0 child c:

```
% relto
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).

% sechk
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- alpha_2(A), a_val_1(A).
c_alpha_2(A) :- alpha_1(A), a_val_1(A).
```

Both theories use the first a-predicate as the outer gate. `relto` later reaches mechanism parent b, whereas `sechk` closes an assumption chain around a. On H3 child c, `relto` reaches b, d, while `sechk` uses only the first, isolated variable a.

Trace-supported explanation. After the failed post-fold entailment check, `relto` first searches for a relative assumption. A KO can return control to folding and admit later BK literals. `sechk` instead generates a provisional assumption and enters `gen5`/`gen6`. Under brave semantics on these cells, the resulting assumption chain closes before the later folds reached by `relto`.

Every H7a `sechk` delta therefore cites only its first BK predictor block. This is not always variable a: for target a, the first predictor is b. Byte-identical H0/H3 `sechk` root deltas consequently reflect a shared learner template, not a shared causal role.

Evidential boundary. H7a does not establish that `sechk` is worse than `relto`, or that first-BK closure is universal. It records a bounded brave-nd interaction on seven exact input-matched pairs.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h7a_relto_vs_sechk.md`

5.8.2 H7b: cautious nd

Question and purpose. H7b asked whether the H7a `relto`/`sechk` difference transfers unchanged when both arms use cautious rather than brave learning.

Controlled comparison. H0 and H3 fixtures, frozen $n = 30$ tables, targets, examples, BK, nd folding, `any` selection, `all` space, and folding ceiling were fixed. Locked `baseline_cautious`/`relto` was compared with experimental `nd_cautious_sechk`/`sechk`. The comparator was not ECAI: both arms used cautious semantics.

Seven targets were run, but the evidence has unequal strength:

Question	Cell	Outcome pair	Evidential use
Q1	H0 c	solved / solved	Comparable deltas and traces.
Q2	H3 c	solved / solved	Comparable deltas and traces.
Q3	H0 b	no solution / no solution	Comparable failed-search cost.
Q4	H3 b	timeout / timeout	Non-comparable: both logs empty.
Q5	H3 a	timeout / timeout	Non-comparable: both logs empty.

Emphasised observation. Both cautious policies solve child c on both tables, but their paired deltas differ. For H0 c:

```
% baseline_cautious / relto
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).

% nd_cautious_sechk / sechk
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- alpha_2(A), a_val_1(A).
c_alpha_2(A) :- b_val_1(A).
```

`sechk` still inserts a new assumption layer around the first a-predicate, but cautious checking prevents the brave early closure and search eventually reaches b.

For H3 c, both policies retain the isolated-a gates and cite mechanism parents b, d somewhere in their contrary structures. The `sechk` theory is thicker:

Arm	Assumptions	Trace lines	Trace character
`relto`	3	257	Relative KO then a lean path to b, d.
`sechk`	5	899	Deeper nests and repeated NEW/`gen5`/KO cycles.

For H0 root b, neither cautious arm emits a delta. The `sechk` trace has approximately 13,818 lines, compared with 5,118 for `relto`, because it makes many more provisional-assumption and `gen5` attempts. H3 targets a and b time out in both arms with empty stdout files; those matched timeout labels reveal no trace-level similarity.

Trace-supported explanation. The same procedural fork as H7a occurs after failed post-fold entailment, but cautious semantics changes which provisional repairs survive. Relative reuse often receives a cautious KO and can expose another fold. The `sechk` path first adds a provisional assumption; when cautious checking rejects an early closure, further nesting and backtracking can eventually reach later b or d predicates. Thus H7b retains the H7a assumption-policy intervention while demonstrating that its observable effect is conditioned by brave versus cautious entailment.

Evidential boundary. The child theories remain non-canonical and, on H3, distractor-retaining. Mechanism-parent occurrence in a contrary is not mechanism recovery. The failed-root comparison concerns cost, and the empty-log H3 timeouts are non-comparable.

Primary evidence record: `docs/experiments/qualitative/M13-C3-binary-collider-and/h7b_cautious_relto_vs_sechk.md`

6 Findings

6.1 Finding 1: Greedy solutions retain all represented BK predictors

6.2 Finding 2: nd solutions are organised around the first BK predictor

6.3 Finding 3: brave nd can solve predictor-ambiguous root targets

6.4 Finding 4: cautious-nd failed-search cost grows with the folding-token ceiling

6.5 Finding 5: cautious-nd trace size grows with grounded sample size

6.6 Finding 6: `relto` and `sechk` produce different assumption-introduction paths

7 Implications